

CHIPPEWA VALLEY, WI, USA

WMO: 726435

Lat: 44.867N

Lon: 91.488W

Elev: 885

StdP: 14.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14991

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.6	-8.5	-21.5	1.7	-12.4	-16.5	2.3	-7.5	21.0	18.6	19.0	19.0	4.8	250	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	89.8	73.0	86.6	71.0	83.9	69.3	75.9	85.2	74.0	83.0	72.1	80.7	11.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	126.3	81.6	71.0	118.0	79.3	69.1	110.3	77.3	39.9	85.3	38.1	82.9	36.3	80.8	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	18.2	16.4	-19.9	94.9	6.6	3.7	-24.7	97.6	-28.5	99.7	-32.3	101.8	-37.1	104.4
			-20.3	79.1	6.5	2.3	-24.9	80.8	-28.7	82.2	-32.3	83.5	-37.0	85.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.2	14.8	18.9	31.4	45.3	57.4	67.3	71.7	69.3	61.7	48.2	34.1	20.9
	DBStd	22.10	13.01	12.52	12.23	9.93	8.40	6.81	5.77	5.59	8.33	8.86	10.27	11.91
	HDD50	4317	1092	872	588	202	26	0	0	0	9	142	484	902
	HDD65	7810	1557	1292	1041	593	264	52	11	22	160	524	927	1367
	CDD50	2573	0	0	13	61	255	520	671	599	359	88	7	0
	CDD65	592	0	0	1	2	28	123	217	156	60	5	0	0
	CDH74	5628	0	0	5	49	372	1185	2087	1327	553	50	0	0
	CDH80	1709	0	0	0	10	105	357	723	359	148	7	0	0
	WSAvg	7.4	7.3	7.6	7.7	8.7	8.2	7.2	6.5	6.1	6.9	7.6	7.6	7.2
	PrecAvg	30.7	1.0	0.7	1.7	3.0	4.1	4.3	3.5	4.4	3.6	2.2	1.5	1.1
(15) Precipitation	PrecMax	45.4	3.5	1.8	4.2	6.1	9.2	7.5	6.9	11.7	8.8	5.6	5.4	2.6
	PrecMin	17.2	0.1	0.0	0.2	1.0	1.2	1.6	0.8	0.3	0.2	0.3	0.1	0.3
	PrecStd	6.7	0.9	0.5	0.9	1.5	1.8	1.5	1.8	2.6	2.1	1.4	1.1	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	51.6	70.9	80.4	88.8	92.5	95.6	91.6	89.3	79.9	66.3	51.6
		MCWB	38.2	45.1	57.5	61.6	68.0	73.7	75.7	73.9	72.0	64.9	54.6	47.0
	2%	DB	38.5	44.3	62.0	72.8	82.7	87.7	90.5	86.8	84.3	73.6	59.6	44.0
		MCWB	35.0	39.6	52.7	55.8	64.5	71.3	74.2	72.1	69.8	62.3	51.7	39.5
	5%	DB	35.0	39.4	55.4	67.8	78.0	84.0	86.9	83.8	80.3	68.3	54.2	39.2
		MCWB	33.0	35.9	46.7	52.1	62.0	68.4	72.2	70.1	67.4	58.3	47.2	36.0
	10%	DB	32.5	35.8	48.3	62.9	73.4	80.8	83.9	81.2	76.5	63.5	49.5	35.9
		MCWB	30.7	32.8	42.2	49.7	59.3	67.0	70.1	68.6	65.2	55.5	43.9	33.4
	0.4%	WB	39.1	44.9	60.5	63.2	72.5	76.7	80.0	77.6	74.6	69.8	57.7	47.8
		MCDB	43.0	50.8	67.8	75.1	84.8	87.9	89.9	86.7	84.0	75.1	62.7	50.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.4	40.1	54.3	58.3	67.6	73.8	76.9	74.9	71.8	63.7	52.5	40.1
		MCDB	37.8	43.8	60.8	69.2	78.1	83.9	86.4	82.9	80.6	70.9	58.3	42.7
	5%	WB	33.2	36.1	47.8	54.5	64.3	71.4	74.6	73.0	69.5	59.9	47.8	36.1
		MCDB	35.0	39.2	54.5	64.2	73.8	80.7	83.8	80.6	77.3	66.3	53.4	38.9
	10%	WB	30.7	33.0	42.2	50.9	61.6	69.1	72.3	70.9	67.2	56.0	44.1	33.6
		MCDB	32.6	35.6	47.8	61.7	70.9	77.7	81.2	78.2	74.6	62.6	49.3	35.7
(35) Mean Daily Temperature Range	5% DB	MDBR	15.8	17.3	18.2	21.2	22.2	20.8	21.0	20.5	21.2	19.0	15.6	14.4
		MCDBR	16.1	18.8	25.5	30.3	28.5	24.7	23.7	22.9	24.9	26.6	23.7	18.1
		MCWBR	13.8	14.9	16.3	16.5	13.3	11.0	10.3	10.7	12.4	15.4	16.1	14.5
		MCDBR	15.7	17.8	23.5	25.8	23.4	20.8	20.9	19.7	21.0	22.2	21.5	16.5
	5% WB	MCWBR	13.7	14.3	15.8	16.0	12.6	11.0	10.5	10.8	11.6	14.8	16.0	13.5
(40) Clear-Sky Solar Irradiance	taub		0.256	0.282	0.315	0.354	0.389	0.391	0.380	0.388	0.359	0.321	0.290	0.262
	taud		2.395	2.317	2.367	2.358	2.301	2.328	2.373	2.364	2.426	2.515	2.510	2.440
	Ebn at Noon		278	288	292	288	279	278	280	274	273	272	263	264
	Edh at Noon		23	30	33	37	40	39	37	36	31	25	21	20
(44) All-Sky Solar Radiation	RadAvg		547	896	1244	1443	1689	1873	1954	1641	1287	817	513	382
	RadStd		56	80	129	150	131	147	80	109	86	101	49	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page